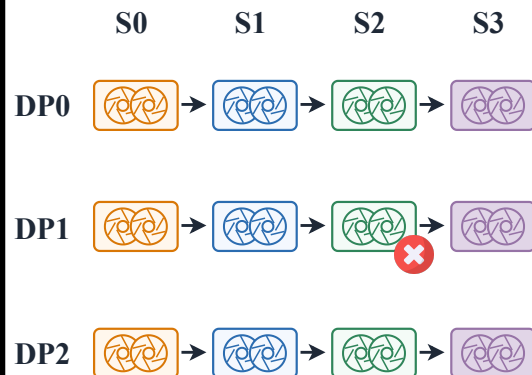
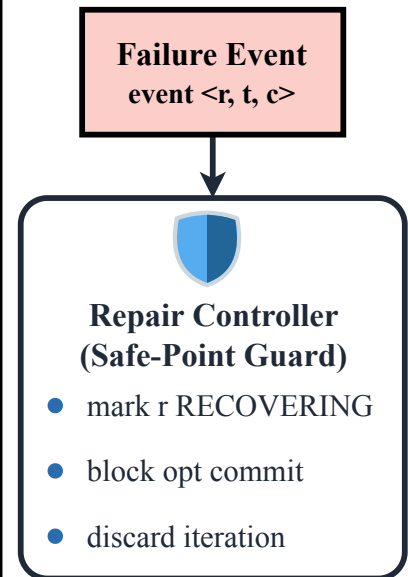


1 Training Job

Sparse MoE Distributed Training (3 DP × 4 Pipeline)

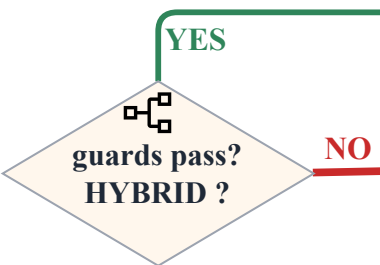
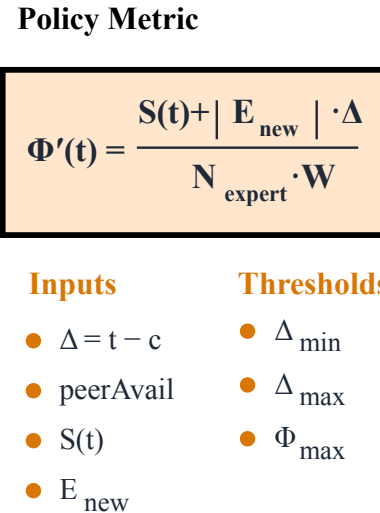


2 Failure Detection + Safe-Point Repair (R1)

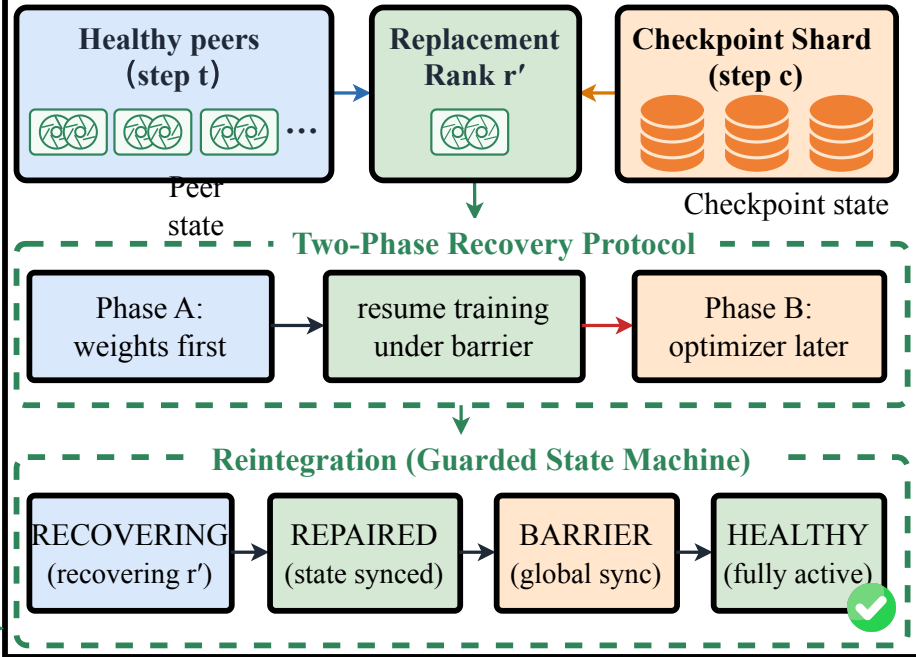


r : failed rank
 t : failure step (current)
 c : checkpoint step

3 Guarded Policy (R2)



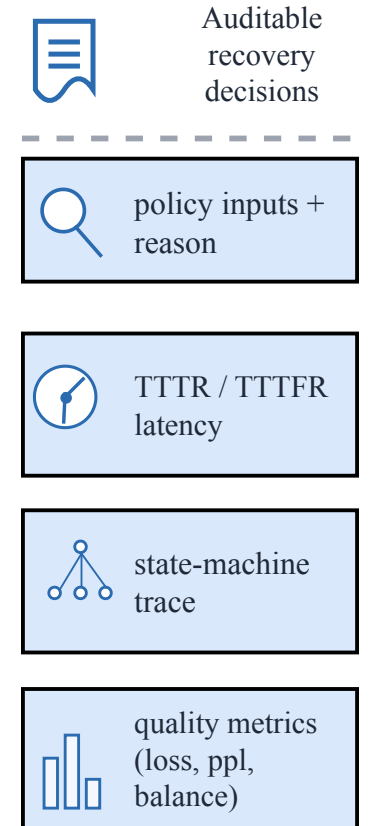
4a Hybrid Recovery (fast path)



4b Checkpoint Restart (fallback)



5 Observability Logs (R3)



Control Flow

Path P (Peer)

Path C (Checkpoint)

Hybrid Path

Restart Path